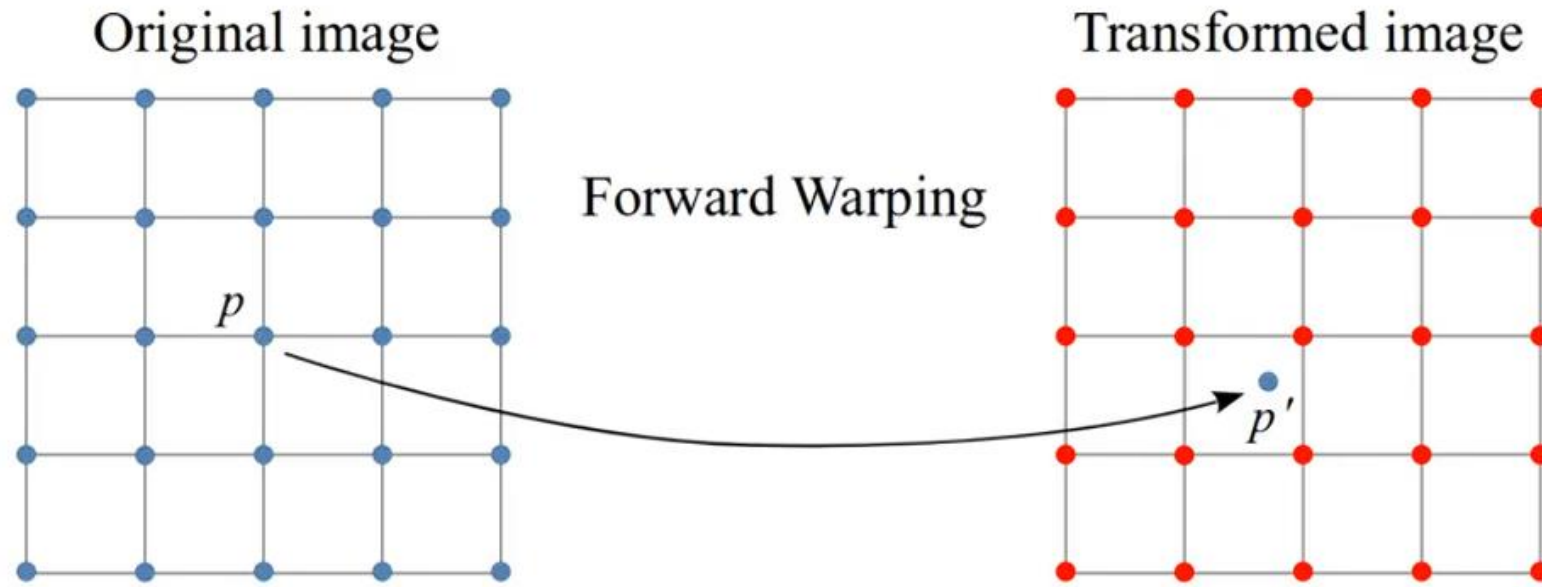


FORWARD AND BACKWARD IMAGE WARPING

Computer Vision with Hüseyin Özdemir

Forward Warping



T is a transformation

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

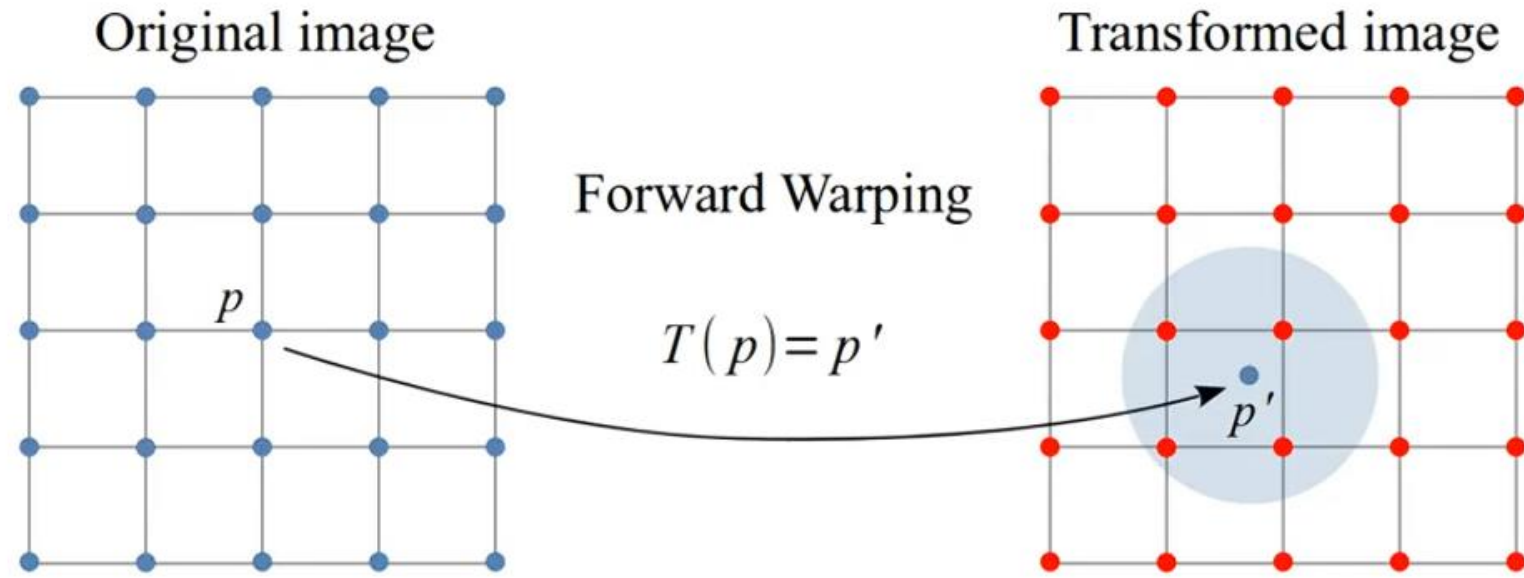
$$T(p) = p'$$

Pixel coordinates are mapped from original image to transformed image

Most of the time transformed coordinates don't
correspond to pixel coordinates

Values of transformed pixels are distributed to neighbor
pixels with a method called Splatting

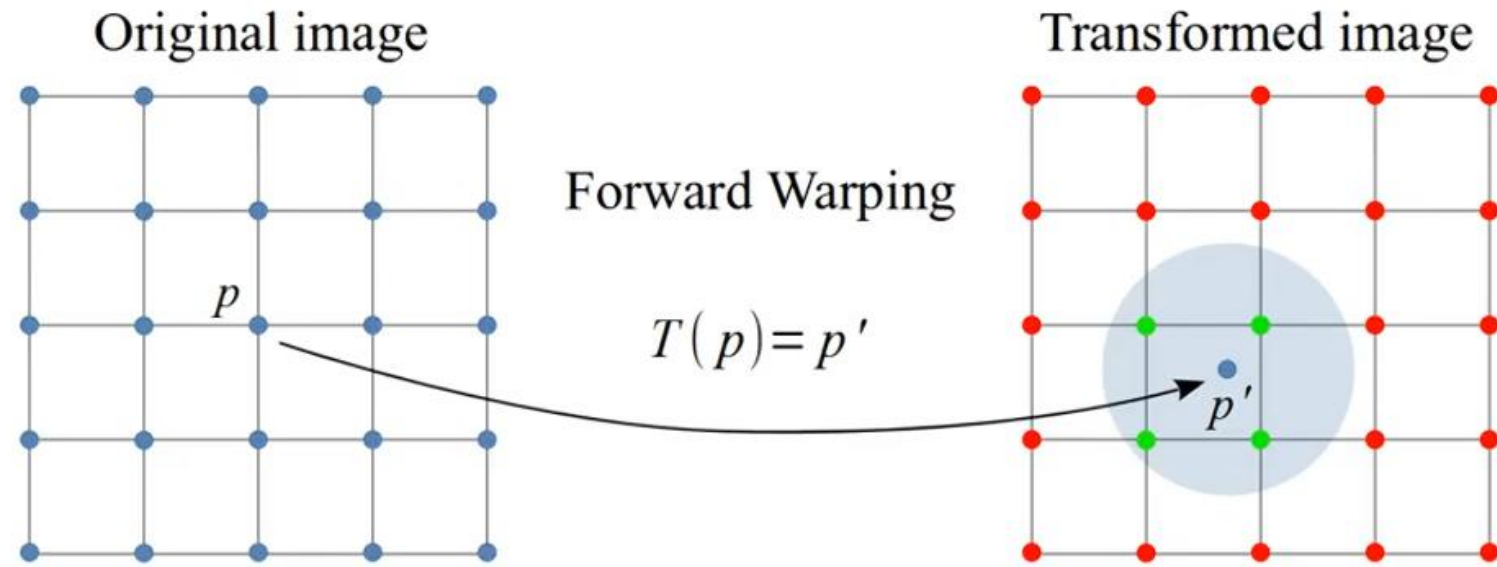
Splatting



Pixel coordinates at p are transformed to p'

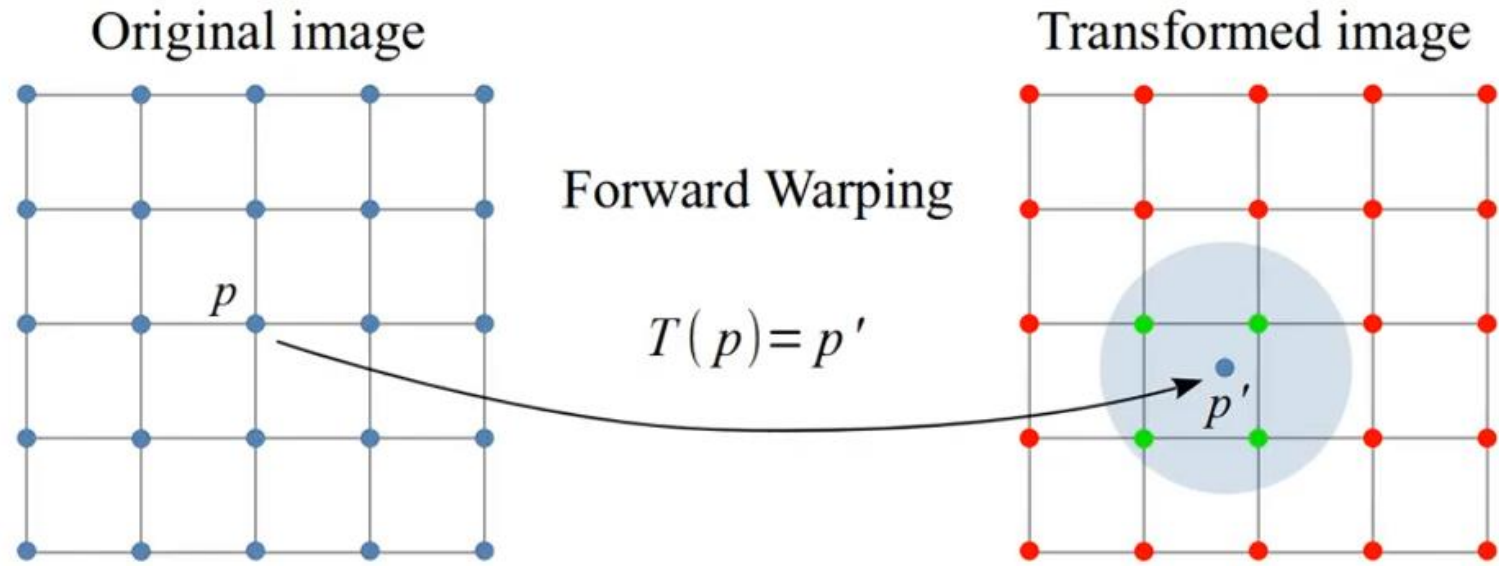
A kernel is placed centering on transformed coordinates p'

Kernel size is adjustable



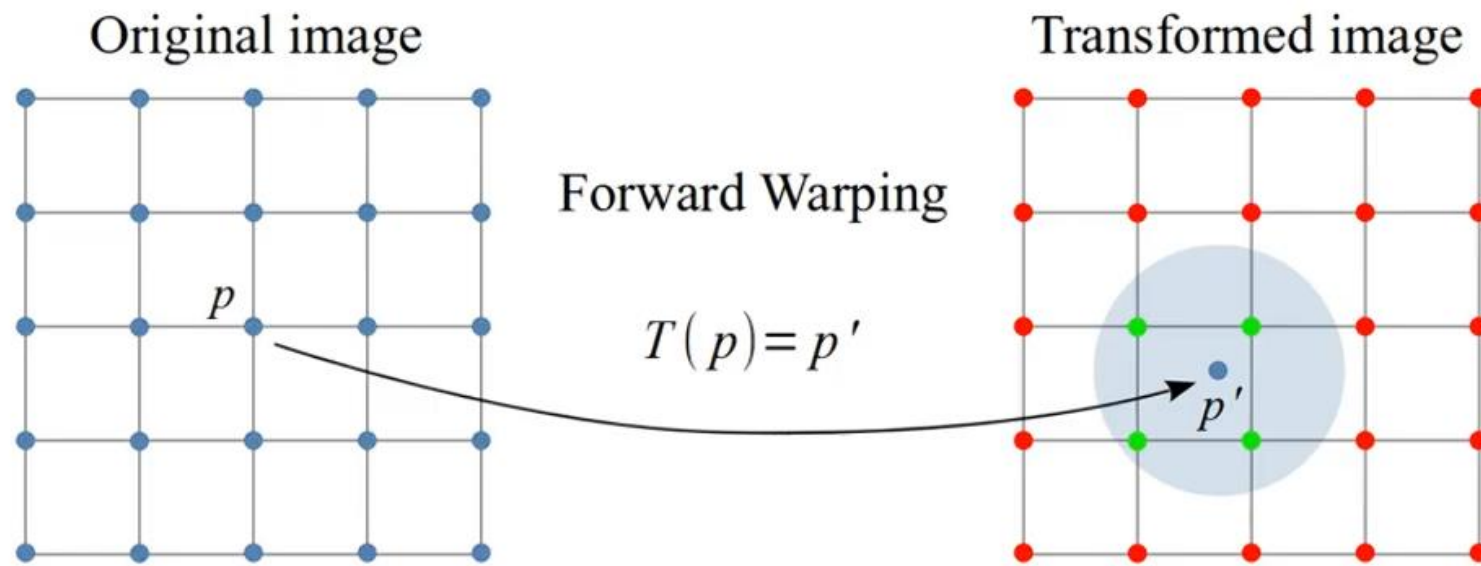
A weight is assigned to each neighbor pixel ● using kernel

Weight decreases as distance from transformed coordinates p' increases



Value assigned to each neighbor pixel is equal to weight
times transformed pixel value

$$assigned_value = weight \times pixel_value$$

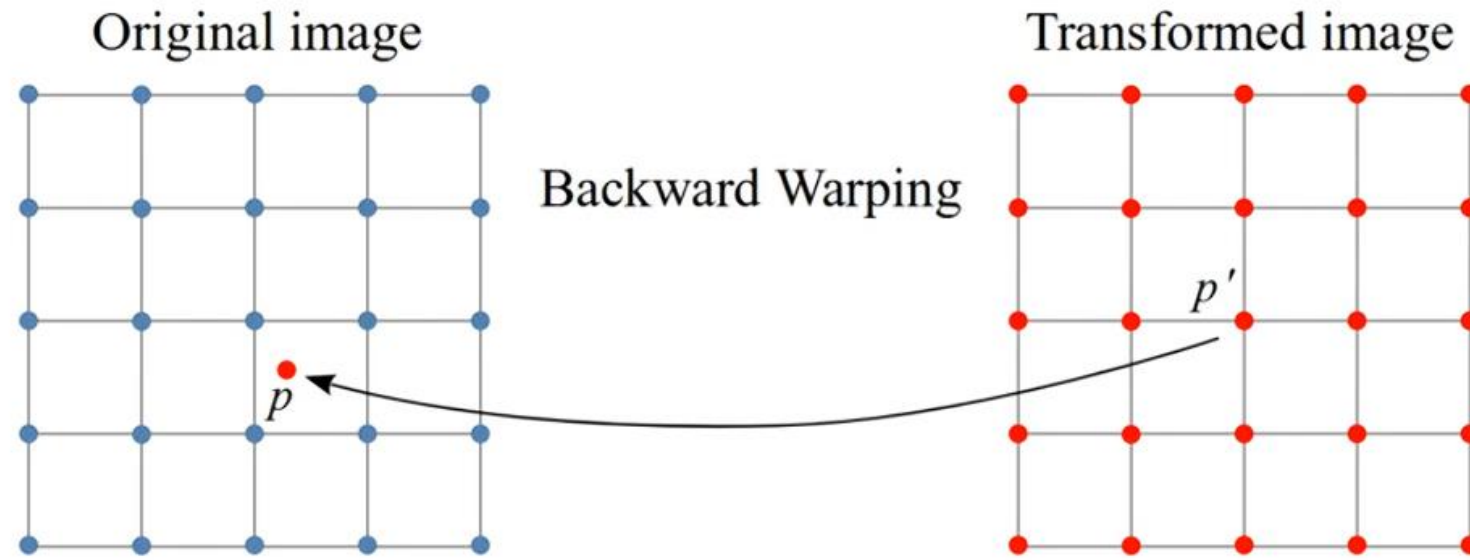


Sum of assigned values and sum of related weights for each pixel of transformed image are stored

Sum of assigned values are divided by sum of weights

This procedure is Weighted Averaging

Backward Warping



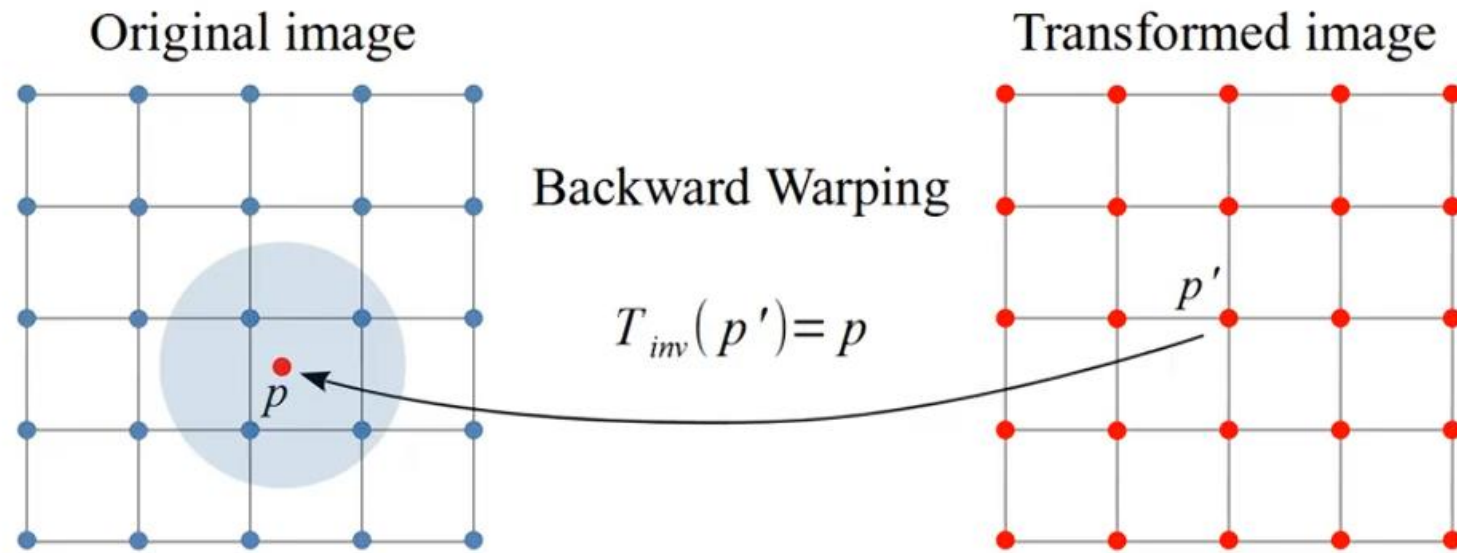
T_{inv} is the inverse transformation $T_{inv}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T_{inv}(p') = p$

Pixel coordinates are mapped from transformed image to original image using inverse transformation

Generally, transformed coordinates don't
correspond to pixel coordinates

Pixel values are interpolated from neighbor
pixels of original image

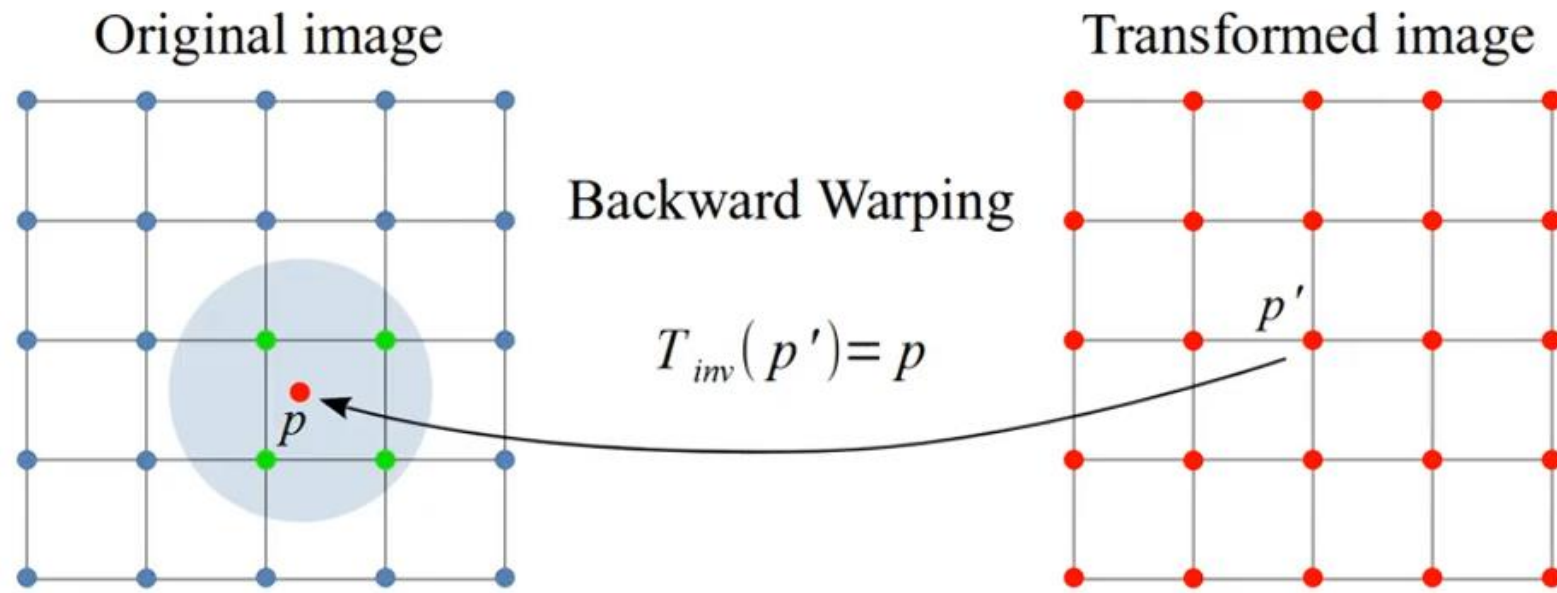
Interpolation



Pixel coordinates at p' are transformed to p

An interpolation kernel is placed centering on coordinates p

Kernel size is adjustable



Pixel value at p is interpolated using neighbor pixels ●

Forward Warping vs. Backward Warping

Forward Warping requires an additional buffer to store
sum of weights assigned to each pixel

Backward Warping is the preferred method if
inverse transformation exists

Reference

- https://www.youtube.com/watch?v=9iN-dAKqcwM&list=PLcEc12xiYZ7r24dJWP4GYwwXUgGS-wl4_&index=8